

Syed Hassnain Abbas

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PROFESSIONAL SUMMARY

Final-year Software Engineering student at FAST-NUCES with 1.5+ years of industry experience building production-grade AI systems, real-time data pipelines, and agentic automation platforms. Currently a Software Engineer at Thunder Energy Intelligence Platform and Co-founder of TACSIGHT, with a track record of shipping end-to-end solutions from architecture to deployment.

EDUCATION

BS Software Engineering

2023 – Expected June 2027

FAST – National University of Computer and Emerging Sciences, Islamabad | CGPA: 2.81/4.0

PUBLICATIONS

Evaluating Feature Extraction Robustness for Morphologically Complex Scripts under Extreme Data Scarcity: A Study on Dot-Dependent Character Recognition | Submitted, 2026

- Muhammad Saood Sarwar, Shahzaib Ur Rehman, **Syed Hassnain Abbas (3rd author)** — ICET 2026, 21st International Conference on Emerging Technologies

Beyond the Horizon: High Precision Aerial Object Classification using YOLO11 Architecture | Submitted, 2026

- Muhammad Saood Sarwar, **Syed Hassnain Abbas (2nd author)** — MCETS 2026, Mediterranean Conference on Emerging Technologies and Systems

Explainable AI for Synthetic Scene Detection: Morphological Artifact Analysis and Forensic Decay Testing Across 192 Categories

| Submitted, 2026

- Muhammad Saood Sarwar, **Syed Hassnain Abbas (2nd author)**, Syed Muhammad Ali Musa — MCETS 2026, Mediterranean Conference on Emerging Technologies and Systems

PROFESSIONAL EXPERIENCE

Thunder Energy Intelligence Platform

Islamabad, Pakistan

Software Engineer

April 2026 – Present

- Built a real-time IoT fuel-theft detection system using time-series anomaly detection and automated alarm lifecycle management, cutting undetected theft incidents by **60%** and enabling rapid field response.
- Rebuilt the **Dynamic Load Shedding engine** with a config-driven architecture and automated data-retention policy, eliminating false **24-hour** outage reports and cutting stale historical data by **~95%** across **Telenor's 6,000+ sites**.
- Optimized the **Asset Discovery pipeline** securing hardware integrity across **Telenor's 6,000+ sites**, fixing core defects in its equipment-history logic and automating daily hardware fingerprinting, movement, and duplicate-device detection fleet-wide.
- Optimized Thunder's **solar forecasting model** with a physics-based recalibration (panel angle, heat, dust, wiring losses), making it **~9%** more accurate than the formula it replaced across Telenor's hybrid tower sites.
- Engineered the **Solar Curtailment Policy and Solar Impact pipelines** powering Thunder's live **Impact Meter** dashboard, tracking **\$99.7K** in fuel savings, **99.3K liters** of diesel, and **270.1 tons** of CO2 reduction across **408 sites**.
- Engineered the **DQI (Data Quality Index) engine**, scoring telemetry across **6 quality dimensions** and **9 automated metrics** per site to catch bad data before it reaches downstream pipelines.
- Built an AI-assisted Selenium scraper extracting IoT device schemas from Teltonika's firmware portals, feeding Thunder's device-config layer and cutting manual schema mapping by **70%**.

AI & Data Engineer Intern

June 2025 – April 2026

- Engineered a multi-protocol IoT management API (Huawei FusionSolar, Growatt, Vertiv) for remote telemetry and configuration across **6,000+ telecom sites** at **98%** data accuracy.
- Built a Kafka-based automation service for Vertiv power systems handling authenticated remote configuration with rollback support, achieving a **98%** success rate across distributed sites.
- Scaled ETL pipelines processing **80k+** daily IoT/energy records with automated validation, maintaining **100%** data integrity across Thunder's analytics stack.

- Built a RAG-based document intelligence platform (**RAGAnything, Qdrant**) turning hundreds of pages of technical manuals into an instantly queryable conversational knowledge base.
- Enhanced Thunder's FESCO energy chatbot with Opik-powered LLM observability and intelligent SQL/RAG query routing, cutting hallucinations by **35%**.
- Built a CrewAI multi-agent pipeline that auto-generates technical documentation from live dashboards, cutting manual UI analysis effort by **75%** and boosting client understanding by **70%**.

Applied ML Intern

January 2025 – May 2025

- Built time-series forecasting models (ARIMA, SARIMA, Prophet, LSTM) predicting load-shedding hours at **93%** accuracy across **6,000+ Telenor telecom sites**.
- Delivered a full-stack analytics web app consolidating battery analysis, power monitoring, tower performance, and forecasting into a single platform.
- Designed interactive Plotly dashboards enabling non-technical stakeholders to visualize energy forecasts and act on them in real time.

Junior Software Engineer Intern | Heavy Mechanical Complex (HMC)

June 2024 – August 2024

Islamabad, Pakistan

- Developed Java desktop applications using MVC, Factory, and Observer design patterns with layered architecture and authentication.
- Applied ML algorithms (KNN, K-Means, Random Forest, Regression) to medical and real-estate datasets, achieving **87%** accuracy in disease and price prediction.
- Worked with BigQuery SQL for large-scale enterprise data querying, transformation, and analysis.
- Built **3+** Power BI dashboards (Comparative Analysis, Sales, P&L) for HMC and Capital Movers and Builders, turning fragmented data into executive insights and improving cost visibility by **15%+**.

System Engineer Intern | Unique Systems

June 2024 – August 2024

Islamabad, Pakistan

- Built a physical server from scratch — installed VMware ESXi, provisioned VMs via vCenter, and deployed it into a live production environment integrated with existing infrastructure.

PROJECTS

Punjab Health Connect (PHC) | Government of Punjab, Pakistan (PHIMC) 2026

- Software Engineer on a live Government of Punjab (PHIMC) healthcare platform connecting diaspora doctors with public hospitals for short-term clinical deployments, covering the full doctor lifecycle from registration to payment.
- Built three role-based portals on **Odoo 18** (Doctor, Hospital, Staff/Admin) spanning **9 user roles** and **10+ backend modules**, translating SRS requirements into production code.
- Designed an Executive Dashboard (Chart.js/ApexCharts, Leaflet maps, Excel/PDF export) auto-refreshing every **5 minutes** for PHIMC leadership.
- Implemented hospital matching via a weighted scoring engine, contract and consent management, and QR-based attendance and remuneration tracking.
- Engineered RBAC authorization, OTP/MFA, and audit logging backed by **PostgreSQL, Redis, and MinIO** — containerized with **Docker** and deployed on PITB's **Kubernetes** infrastructure.
- Set up a **GitHub Actions** CI/CD pipeline to auto-build, migrate, and deploy Odoo updates on every push.
- Wrote **18+** unit and integration test modules covering SLA compliance, credential workflows, and contract/attendance logic for a live multi-hospital rollout.

Hisaab – AI Financial Copilot

- Built a full-stack AI financial assistant for Pakistani shopkeepers supporting multi-modal input — voice (Urdu, Roman Urdu, English), receipt/invoice image capture, and manual entry.
- Powered by a conversational chatbot with custom **MCP** tools for natural-language business queries, paired with a real-time inventory system for stock tracking and alerts.
- Integrated an AI Advisor and autonomous goal-tracking agent that nudges shopkeepers back on track when performance dips, driving a **40%+** improvement in daily profit tracking.

EduBot – Intelligent Learning Assistant

- Built an AI study platform (RAG + custom MCP framework) unifying course PDFs, lecture videos, and notes into one queryable knowledge base, driving an **80%** improvement in exam-prep efficiency.
- Automated lecture summarization (YouTube to structured notes) and Jira task creation from study sessions, cutting content review from hours to minutes.

VisionAI – Multimodal Accessibility Assistant | *Internship Project*

- Built a real-time multimodal AI assistant (computer vision + LLMs) guiding non-technical and elderly users through digital systems via natural voice and text responses.
- Adoption studies showed a **65%** improvement in independent system usage among non-technical users, reducing dependence on support staff.

VoxAssist – Personal Voice-Clone Assistant | *Personal Project*

- Built a voice-clone assistant that turns a **10-20 second** recording into a natural-sounding clone, reading back notes, reminders, and digests in your own voice.
- Got a **2-billion-parameter** voice model running fully offline on CPU with no GPU or cloud service, working around a Windows security block — keeping voice data fully private.

TellScape – AI Web-Scraping Agent | *Automation Tool*

- Built an AI-driven scraping agent that turns a single plain-English instruction into a fully autonomous browsing session, pulling clean structured data with zero hand-written selectors.
- Layered in automatic enrichment that cross-references each result against live public data, returning a decision-ready picture in one shot instead of a flat link list.

SKILLS

AI & ML: AI Agents, RAG Systems, LLM Observability (Opik), Time Series (ARIMA, SARIMA, Prophet, LSTM), ML Models, CNN, SVM, SLM, MCP, VLM, LightRAG, TTS/Voice Cloning (VoxCPM)

Data Engineering: ETL Pipelines, Kafka, Poller Services, SQL/PostgreSQL, BigQuery, asyncpg, Pandas, NumPy, pytest

Domain Expertise: Solar/PV Forecasting (PVsyst methodology), Load-Shedding Analytics, Solar Curtailment & DG Policy Modeling, Telecom Energy Infrastructure

Frameworks & Tools: CrewAI, OpenClaw, Dify, LangChain, LangGraph, Qdrant, FastAPI, Gemini, Groq, Selenium, Playwright, ThreeJS, Docker, Kubernetes, Odoo

Visualization: Plotly, Power BI, Seaborn, Matplotlib

Programming: Python, Java, C++, C, JavaScript

Other: VMware ESXi, vCenter, Git